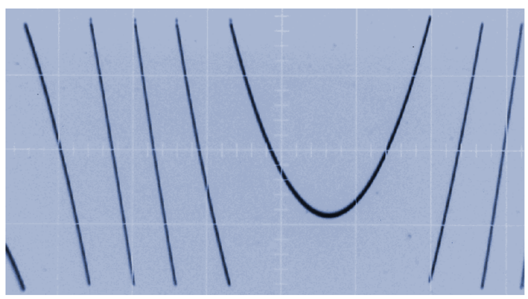
The Teezer, a Sawtooth VCO with Thru-Zero Frequency Modulation

Ian Fritz, March 2010

This module is a sawtooth-based VCO capable of frequency modulation (FM) extending past zero frequency into the negative-frequency regime. Thru-zero FM provides a much wider and richer variety of sounds than "ordinary" (positive frequency only) FM.

The negative-frequency version of a waveform is simply a time-reversed replica of the original waveform. When a VCO is modulated through zero frequency, the waveform slows down to a stop and then speeds back up in the reverse direction. Here is a picture of the core sawtooth wave being modulated to negative frequencies. On the left side of the picture the wave is a down ramp, and on the right side it's an up ramp, with the zero-frequency point being at the broad dip.



The Teezer's output waveforms also include triangle and sine waves, for producing sounds with fewer high harmonics, as typically used for bell sounds, train whistles, and so on. The unit also features a variable synchronization control that can be adjusted over a range of settings from hard sync to a fairly loose soft sync.

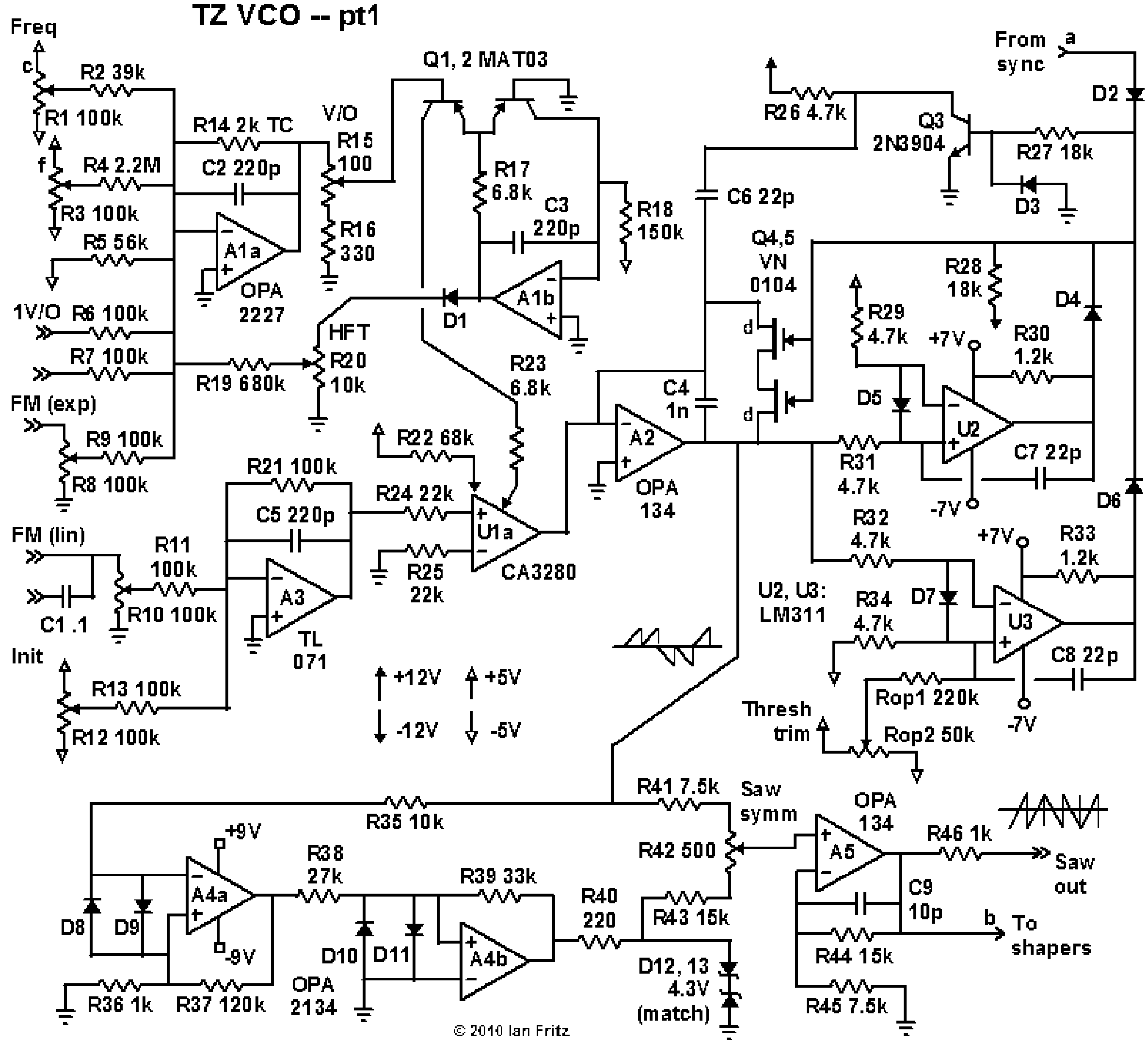
The module can also serve as a highly accurate and stable "ordinary" VCO (i.e., without the deep FM), with upramp, downramp, triangle and sine output waveforms.

Schematic:

The schematic diagram is divided into three parts:

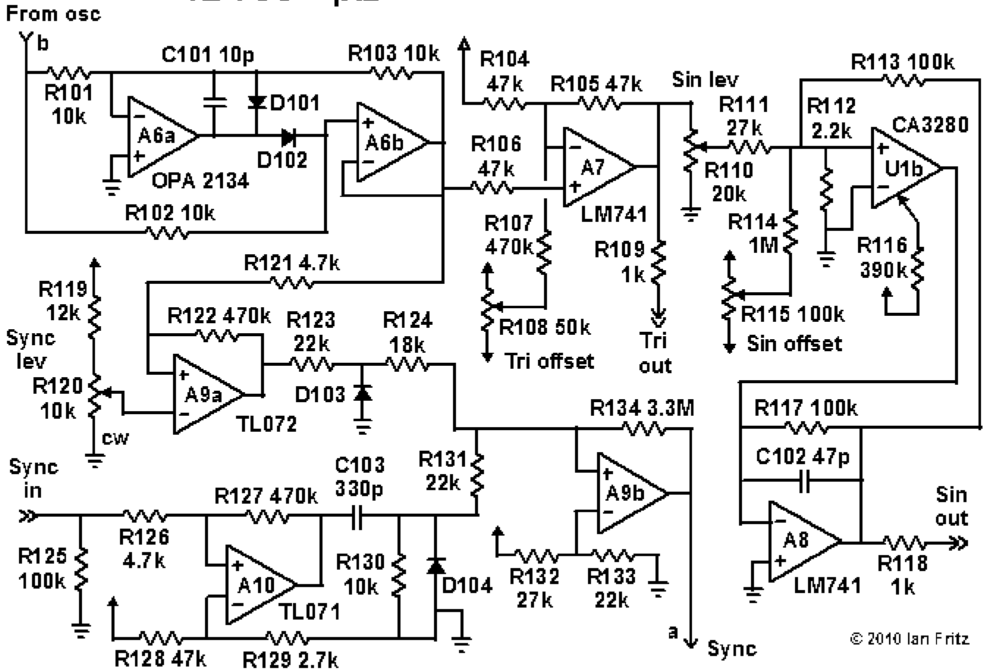
(1) the VCO core and modulation circuitry,
(2) the waveshapers and the sync circuit, and
(3) the power supply decoupling arrangement and the I/O connectors.

TZ VCO -- pt1

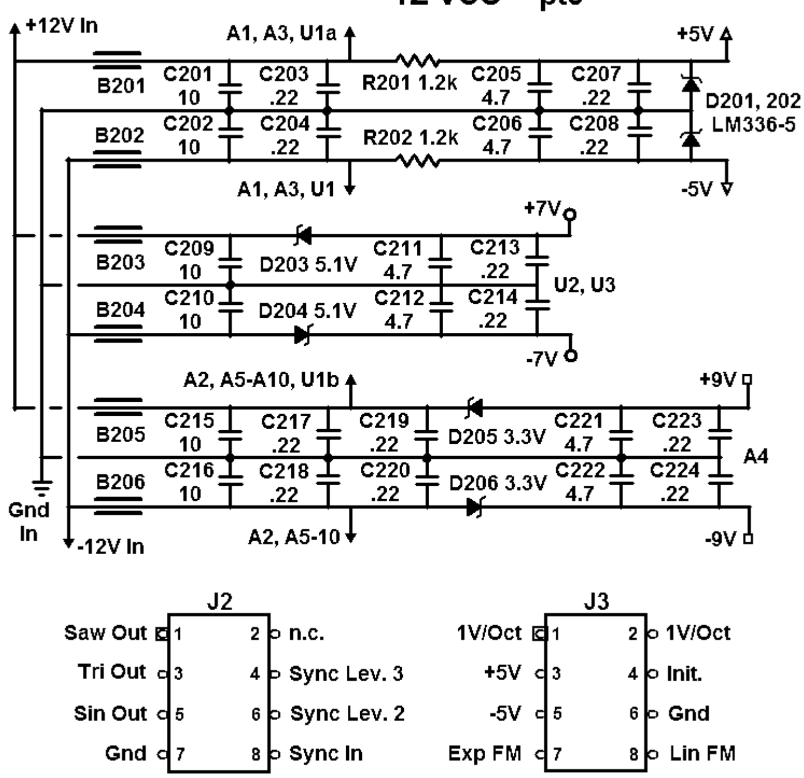


Freq
c
R2 39k
R1 100k
R14 2k TC
V/O
R15
100
f
R4 2.2M
C2 220p
R3 100k
R16
330
R5 56k
A1a
OPA
2227
1V/O
R6 100k
R7 100k
R19 680k
HFT
R20
10k
FM (exp)
R9 100k
R8 100k
Q1, 2 MAT03
R17
6.8k
C3
220p
R18
150k
A1b
D1
R23
6.8k
R22 68k
R21 100k
C5 220p
R24 22k
U1a
CA3280
R25
22k
FM (lin)
R11
100k
C1 .1
R10 100k
A3
TL
071
Init
R13 100k
R12 100k
+12V
+5V
-12V
-5V
C4
1n
A2
OPA
134
R26 4.7k
C6 22p
Q3
2N3904
From
sync
a
D2
R27 18k
D3
Q4,5
VN
0104
R28
18k
R29
4.7k
D4
+7V
R30
1.2k
D5
U2
R31
4.7k
C7 22p
-7V
D6
R32
4.7k
+7V
R33
1.2k
U2, U3:
LM311
R34
4.7k
D7
U3
C8 22p
Rop1 220k
-7V
Thresh
trim
Rop2 50k
R41 7.5k
Saw
symm
R35 10k
OPA
134
A5
R46 1k
Saw
out
R42 500
+9V
R38
27k
R39 33k
A4a
R40
220
R43 15k
C9
10p
b
To
shapers
D8
D9
-9V
D10
D11
A4b
D12, 13
4.3V
(match)
R44 15k
R36 1k
R37 120k
OPA
2134
R45 7.5k

TZ VCO – pt2

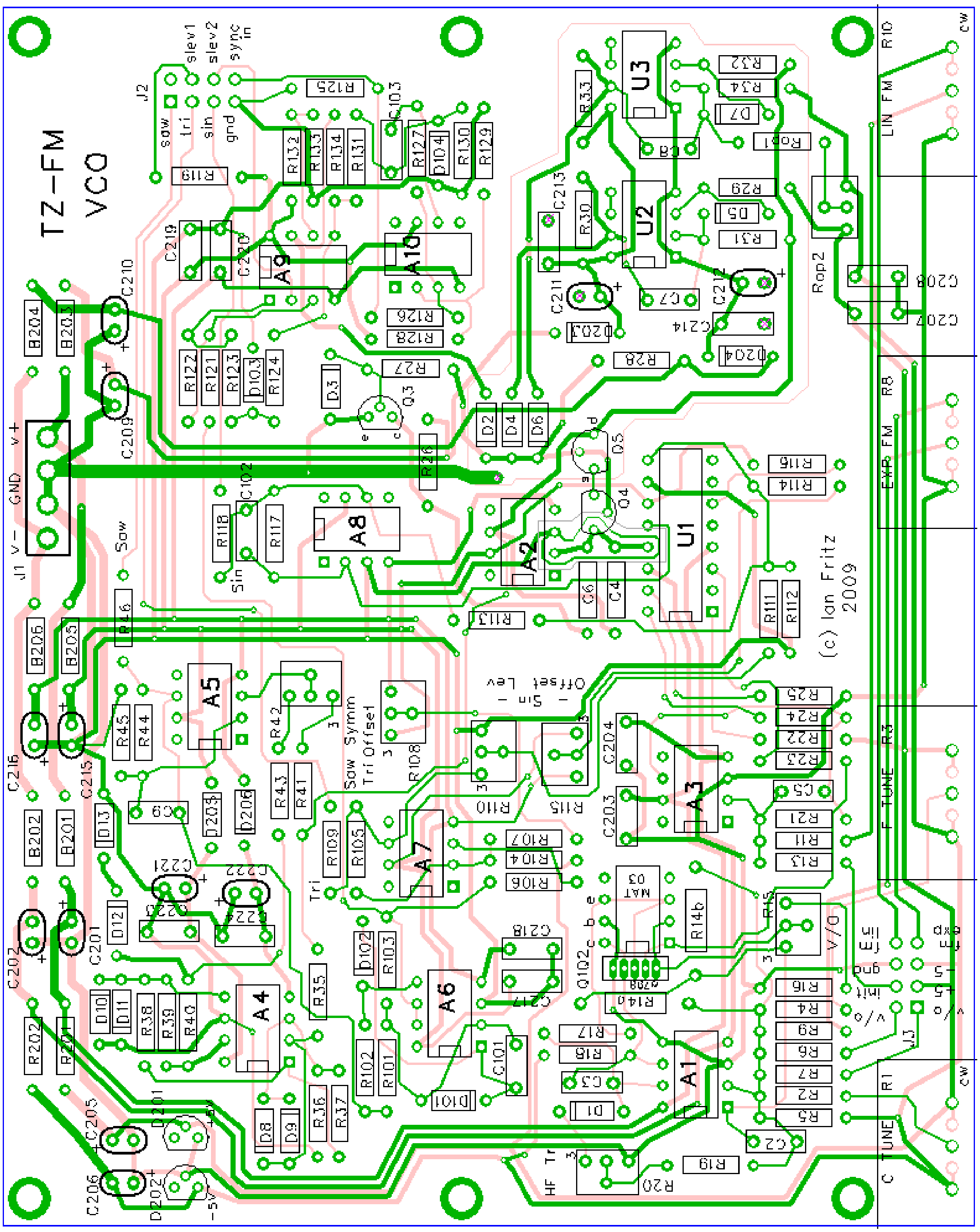


TZ VCO -- pt3



Circuit Boards:

The PCB (MOTM format) is shown below. Construction is straightforward. It is suggested to use shielded cables or twisted pairs for the FM and V/O inputs.



Component List:

ICs:

A1	OPA2227	A7, A8	LM741
A2, A5	OPA134	A9	TL072 (or similar)
A3, A10	TL071 (or similar)	U1	CA3280
A4, A6	OPA2134	U2, U3	LM311

Transistors:

Q1, Q2	MAT03 (2SA798 with reduced performance), matched PNP transistor pair
Q3	2N3904
Q4, Q5	VN0104 (MOSFETs)

Si Signal Diodes(1N4148, etc.):

D1-D11, D101-D104 (15 ea)

Zener/Reference Diodes:

D12, D13	1N5229B, 4.3 V Zener, matched pair
D201, 202	LM336-5.0, 5.0 V reference
D203, 204	1N5231B, 5.1 V Zener
D205, 206	1N5226B, 3.3 V Zener

Resistors (Metal film 1%, 50ppm tempco preferred):

220 Ohm	R40
330 Ohm	R16
1 kOhm	R36, R46, R109, R118
1.2 kOhm	R30, R33, R201, R202
2 kOhm	R14 3300 ppm/K tempco
2.2 kOhm	R112
2.7 kOhm	R129
4.7 kOhm	R26, R29, R31, R32, R34, R121, R126
6.8 kOhm	R17, R23
7.5 kOhm	R41, R45
10 kOhm	R35, R101-R103, R130
12 kOhm	R119
15 kOhm	R43, R44
18 kOhm	R27, R28, R124
22 kOhm	R24, R25, R123, R131, R133
27 kOhm	R38, R111, R132
33 kOhm	R39
39 kOhm	R2
47 kOhm	R104-R106, R128
56 kOhm	R5
68 kOhm	R22
100 kOhm	R6, R7, R9, R11, R13, R21, R113, R117, R125
120 kOhm	R37
150 kOhm	R18
390 kOhm	R116

470 kOhm	R107, R122, R127, Rop1 (optional)
680 kOhm	R19
1 MOhm	R114
2.2 MOhm	R4
3.3 MOhm	R134 (5% carbon film OK)

Pots:

R120	10 kOhm linear taper
R1, R3, R12	100 kOhm lin taper (cermet for best stability)
R8, R10	100 kOhm log taper

Trimpots (cermet):

100 Ohm	R15
500 Ohm	R42
10 kOhm	R20
20 kOhm	R110
50 kOhm	R108, Ropt2 (optional)
100 kOhm	R115

Caps (polystyrene or C0G):

22 pF	C6, C7, C8
220 pF	C3
1.0 nF	C4

Caps (plastic or ceramic):

10 pF	C9, C101
47 pF	C102
220 pF	C2, C5
330 pF	C103
0.1 uF	C1
0.22 uF	C203, C204, C207, C208, C213, C214 C217-C220, C223, C224

Caps (electrolytic):

4.7 uF	C205, C206, C211, C212, C221, C222
10 uF	C201, C202, C209, C210, C215, C216

Ferrite beads:
B201-B206

Building and Calibration:

The circuit was developed and tested with +/-12 V power, and is *not* expected to work properly at +/-15 V. People with +/-15 V supplies can make modifications as discussed below in the Addendum.

Many of the components and component values are critical for proper operation. It is strongly suggested that builders not make any substitutions unless they are certain that circuit operation will not be affected.

Correct accuracy of the output waveforms requires matching the two Zener diodes D12 and D13. To measure the Zener breakdown voltage, the diode should be connected between ground and the +12 V supply with a series resistor (1 kOhm or so) between the supply and the device. The voltage across the diode should be measured for 6 - 10 devices, and the pair with the closest results should be used in the circuit.

Capacitor C1 is to be mounted on the panel, either with its own jack or connected to the DC input jack with a switch.

Verify operation:

(1) Check that there is a correct signal at each of the three outputs: Adjust the "Coarse Tune" ("Freq c" on the schematic) and "Initial Freq" ("Init" on the schematic) controls to get a signal in the audio range. Amplitudes should be near +/- 5 V.

(2) With the "Initial Freq" control at each end of its range, check that the "Coarse Tune" control produces a large frequency swing and that the "Fine Tune" control produces a small one. The overall range should be from lower than 0.01 Hz (100 sec period) to over 30 kHz.

(3) Verify that the varying the "Initial Freq" control brings the frequency to zero near the center of its range, and that upramps are obtained on one side of zero and downramps on the other side. Check that the frequency at the two extremes are roughly equal.

(4) Try the modulation inputs to make sure there is plenty of depth.

Tuning and tracking:

Tuning and tracking are done in a similar manner to that used for "regular" VCOs.

(1) Set the "Initial Freq" control fully to the left.

(2) Adjust the "V/O" trimmer (R15) for a proper octave between 100 Hz - 200 Hz.

(3) Adjust the "HFT" trimmer (R19) to set the octave between 4 kHz and 8 kHz.

(4) Repeat (2) and (3) until both octaves are in tune.

(5) Check the tracking with the "Initial Freq" control fully to the right. Each of the two octaves should be correct to within +/- 0.5%.

Waveform adjustment:

An oscilloscope should be used to adjust the output waveshapes.

(1) Adjust the "Saw symm" trimmer (R43). This is most conveniently done by observing the "Triangle" output and adjusting for the smallest offset at the upper tips of the wave. The offset will be slightly different for positive and negative "Initial Freq" settings, so check both and adjust for the best overall shape. If the accuracy is not good enough, then better matching between the Zeners may be required, or the value of C6 may need to be changed slightly.

(2) Adjust the "Tri offset" trimmer (R108) to center the "Triangle" output signal about zero volts.

(3) Adjust the "Sine" output wave for the purest signal by adjusting the "Sin lev" and "Sin offset" controls (R110 and R115). Once the trimmers are near their correct position, the final adjustment may conveniently done by ear, adjusting the trimmers for minimum amplitude of the second and third harmonics.

(4) (Optional) Very rarely there is a minor waveform glitch stemming from the upramp and downramp times

being slightly different. The trimmer Ropt2 and resistor Ropt1 may be used to fix this. For the initial setup, it is suggested to leave both of these out, or at least set the trimmer to the clockwise end of its range. The waveform anomaly occurs when (a) the "Initial Freq" control voltage is small (<1 V), (b) the modulation frequency and the VCO core frequency are almost exactly the same, and (c) the modulation depth is such that one of the wave segments is at the borderline between turning around (through zero frequency) and resetting. At this point the Saw signal may develop varying subharmonics that sound just like a creaky door. Adjusting the trimmer should eliminate this effect. Note that it requires very careful adjustments of frequency and depth to get this condition to occur. But if it appears to be a problem, just adjust the trimmer until it goes away. This must be done for both positive and negative values of the "Initial Freq" control.

Variable sync operation:

Although no adjustment is needed, the operation of the sync circuit should be checked. With the "Sync Depth" ("Sync lev" on schematic) control fully clockwise, a hard sync should be observed, i.e., the oscillator should always lock to the frequency of the modulation source. With the control near the counterclockwise limit, synchronization should occur only over fairly narrow frequency bands.

Operation:

A few brief words about circuit operation, beginning with the part 1 diagram above. The circuitry around opamps A1a and A1b forms a standard 1 V/Oct exponential converter, with the matched PNP transistor pair Q1, Q2 providing the exponential current output. The trimmers R15 and R20, respectively, set the initial and high frequency tracking.

Rather than feeding an integrator directly, as in a "normal" VCO, the exponential current source drives the control-current input of U1a, a CA3280 OTA. Opamp A3 sums a DC bias voltage (denoted as "Init" or "Initial Freq") with the external modulation signal("FM lin" or "Lin FM"). Its output voltage feeds the bipolar input of the OTA. The OTA's output current is then proportional to the exponential current and to the total input voltage of A3.

The modulated current output of U1a goes to the opamp integrator formed by A2 and C4. Because the control current is bipolar, the integrator can ramp either upward or downward. In either case, when the magnitude of the signal exceeds 5 V, the switch consisting of the MOSFET pair Q4, Q5 discharges the integrating capacitor to 0 V. The comparator U2 drives the discharge switch for an upramp signal, and U3 drives the switch for a downramp. The circuitry around Q3 produces a pulse to cancel the charge injected into C4 by the reset pulses.

The integrator output -- the "raw" sawtooth signal -- does not exhibit the continuous phase reversal required of a through-zero VCO. The circuitry along the bottom of the drawing detects whether the signal is positive or negative and then shifts the positive signals downward and the negative signals upward to produce the proper signal. Trimmer R42 (Saw symm) sets the correct ratio of the raw sawtooth to the square wave produced by A4, D12 and D13, for accurate subtraction of these signals.

In the part 2 drawing the waveshaper and sync circuits are shown. A6a and A6b form a full-wave rectifier, which produces a 0 V to 5 V triangle wave. This signal is shifted and amplified by A7 to produce the final "Triangle" output signal. A sine wave is derived from the triangle by U1b in conjunction with A8. Feedback via R113 improves the purity of the sine signal. The "Sin lev" and "Sin offset" trimmers are adjusted for the best sinusoidal shape.

A9, A10 and associated components provide a variable sync function. Panel control R120 ("Sync lev") sets the capture range of the sync function, from soft sync to hard sync, by detecting the time window where the raw triangle signal is above a variable threshold voltage. A10 acts as a comparator to square up the driving signal ("Sync") so that any signal crossing 0.6 V may be used as the master oscillator. A9b produces a pulse whenever the time of the sync signal's threshold crossing is within the time window defined by the Sync lev" control. This pulse resets the VCO core to 0 V.

The part 3 drawing illustrates the power supply decoupling and sub-regulation scheme. The input power is split into three branches by the three pairs of ferrite beads. The first branch supplies power to the exponential converter and includes sub-regulated 5 V reference sources. The second branch supplies the reset pulse circuitry and includes sub-regulation to speed up and reduce the power consumption in U2 and U3. The third branch supplies the remaining circuitry and includes sub-regulated power to decouple the comparators A4a and A4b.

Addendum: Modifications for +/- 15V Power

The most convenient method of adapting this project to +/- 15 V power is to reduce the power lines to 12 V using series Zener diodes. This may be done without board modification by replacing the six ferrite beads with Zeners. Two different device types are used because the different sections draw different currents. The substitutions are:

B201, B202 1N5225B, 3.0 V Zener diode
B203-B206 1N5223B, 2.7 V Zener diode

Important: use *only* the specified devices.

The following figure shows the location and orientation of the diodes.